PRACTICAL 11
Storage Management

Aim

To study MBR/GPT, partitions, file systems, mounting, UUID, labels, LVM and swap.

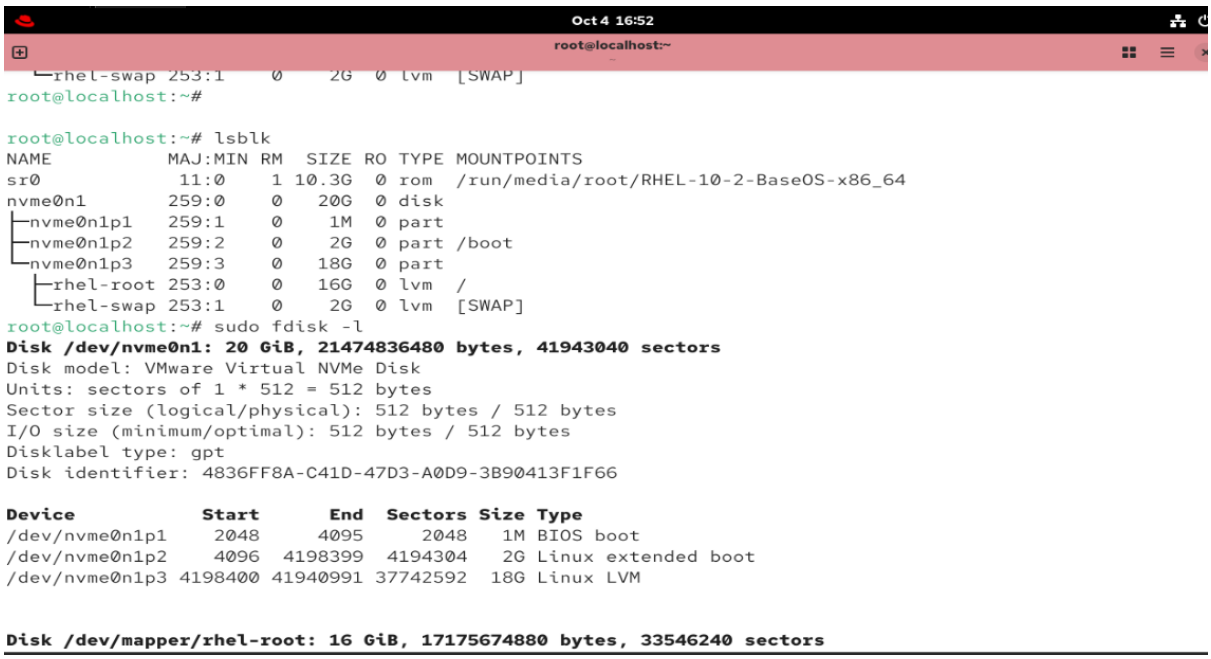
WARNING: Perform this practical only on an additional virtual disk.

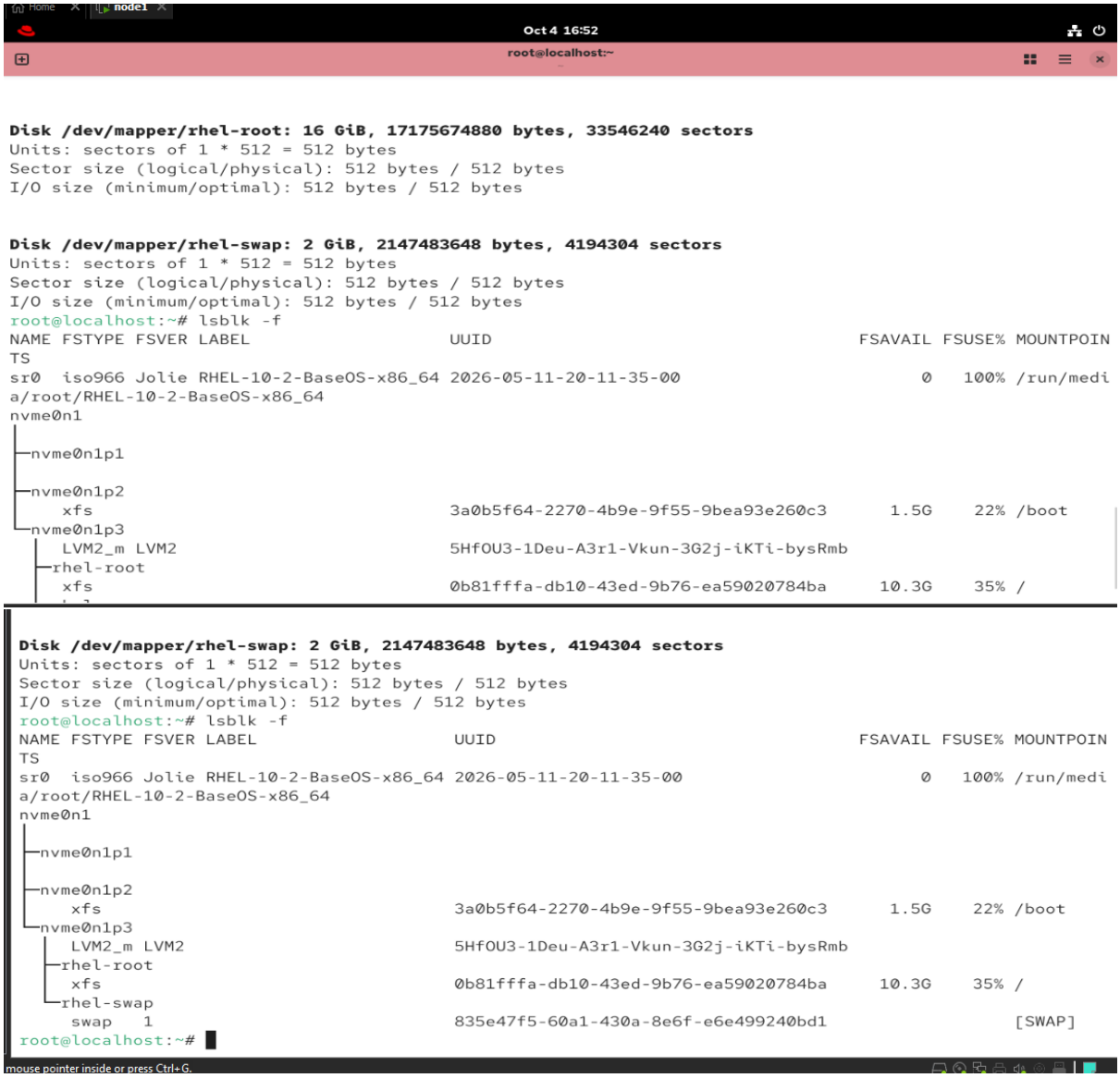
A. Identify disks

lsblk

sudo fdisk -l

lsblk -f





B. MBR/GPT

Check partition table:

sudo fdisk -l /dev/sdb

For GPT information:

sudo parted /dev/sdb print

Do not use /dev/sda unless you have positively confirmed it is the disposable lab disk.

C. Create partition

Example:

sudo fdisk /dev/sdb

Inside fdisk, commands vary according to the desired partition table. Always verify the disk before writing changes.

D. Create XFS

After creating a disposable partition such as /dev/sdb1:

sudo mkfs.xfs /dev/sdb1

E. Create ext4
sudo mkfs.ext4 /dev/sdb1

F. Create VFAT
sudo mkfs.vfat /dev/sdb1

G. Mount
Create mount point:
sudo mkdir /mnt/lab
Mount:
sudo mount /dev/sdb1 /mnt/lab
Verify:
df -h /mnt/lab
Unmount:
sudo umount /mnt/lab

H. UUID
lsblk -f
or:
sudo blkid /dev/sdb1
Example:
/dev/sdb1: UUID="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx" TYPE="xfs"

I. Mount using UUID
Edit:
sudo vi /etc/fstab
Example:
UUID=YOUR-UUID-HERE /mnt/lab xfs defaults 0 0
Test:
sudo mount -a
Then:
df -h
Students must use the **actual UUID produced by their own system**, not the example UUID.